

Seneca

SEG800

Digital Signal and Image Processing

Introduction to Image Processing

Seneca College

Overview

- Introduction to Digital Image Processing
- Human Vision System
- Image Acquisition and Digital Images
 - Neighborhoods and Connectivity
 - Distance measures

Basic Definitions

- **Image:** A two-dimensional function, $f(x, y)$, where x and y are spatial (plane) coordinates, and the amplitude of f at any pair of coordinates (x, y) is called the intensity or gray level of the image at that point.
- **Digital image:** An image for which x , y , and the intensity values of f are all finite, discrete quantities.
- **Pixel:** Each element of a digital image - with a specific (x, y) location and thus $f(x, y)$ intensity. Short for “picture element”.
- **Digital Image Processing:** Processing a digital image with a digital computer.

Processing Levels leading to Related Fields

- **Image Processing:** More focus on low-level processes involving operations such as noise reduction, contrast enhancement, and image sharpening.
 - Input: Image; Output: Image
- **Image Analysis:** More focus on mid-level processes involving tasks such as segmentation, feature extraction, and object recognition.
 - Input: Image; Output: Attributes extracted (e.g., edges, contours, and the identity of individual objects).
- **Computer Vision:** More focus on higher-level processing, “making sense” of an image as in human vision or better.

History

- 1920s: Newspaper industry; transmission of a picture on Bartlane cable (time down to 3 hours; previously more than a week to transport)
- 1960s: Computers powerful enough for image processing
- 1970s: Computerized tomography (CT), X-Ray

Figure 1.1

A digital picture produced in 1921 from a coded tape by a telegraph printer with special typefaces.



Figure 1.2

A digital picture made in 1922 from a tape punched after the signals had crossed the Atlantic twice.



Figure 1.3

Unretouched cable picture of Generals Pershing (right) and Foch, transmitted in 1929 from London to New York by 15-tone equipment.



Figure 1.4

The first picture of the moon by a U.S. spacecraft. Ranger 7 took this image on July 31, 1964 at 9:09 A.M. EDT, about 17 minutes before impacting the lunar surface.



Image Categories according to Source

- Image are created from various energy sources (e.g., X-ray, visual, infrared, and so on).
- The principal energy source for images in use today is the **electromagnetic (EM) energy spectrum**.
- Other important sources of energy include acoustic, ultrasonic, and electronic (in the form of electron beams used in electron microscopy). *Synthetic* images, used for modeling and visualization, are generated by computer.

Figure 1.5

The electromagnetic spectrum arranged according to energy per photon.

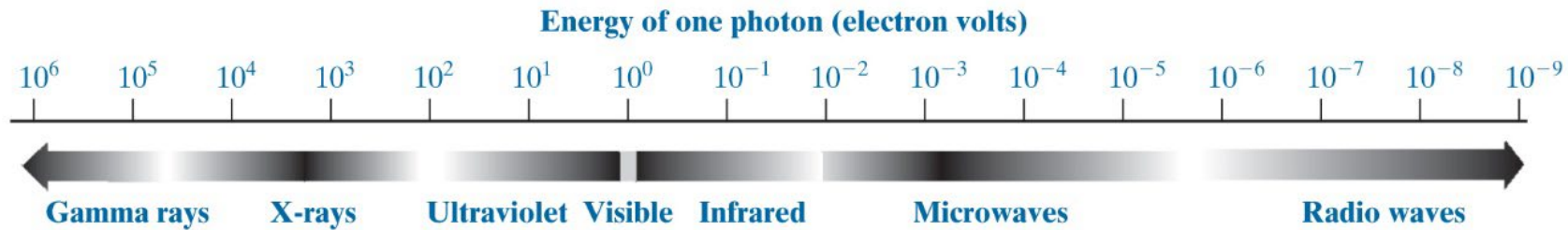


Figure 2.10

The electromagnetic spectrum. The **visible spectrum** is shown zoomed to facilitate explanations, but note that it encompasses a very narrow range of the total EM spectrum.

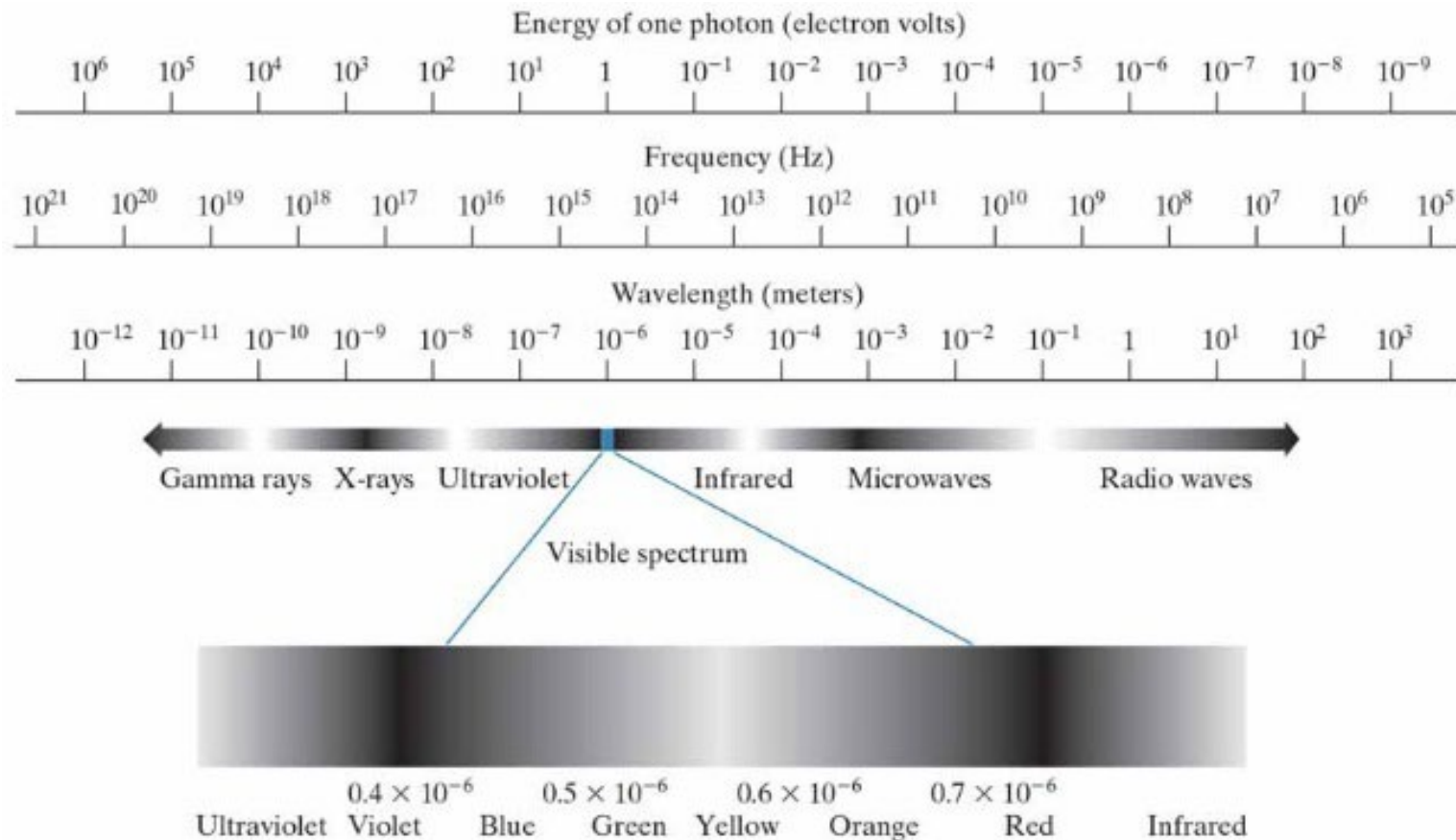
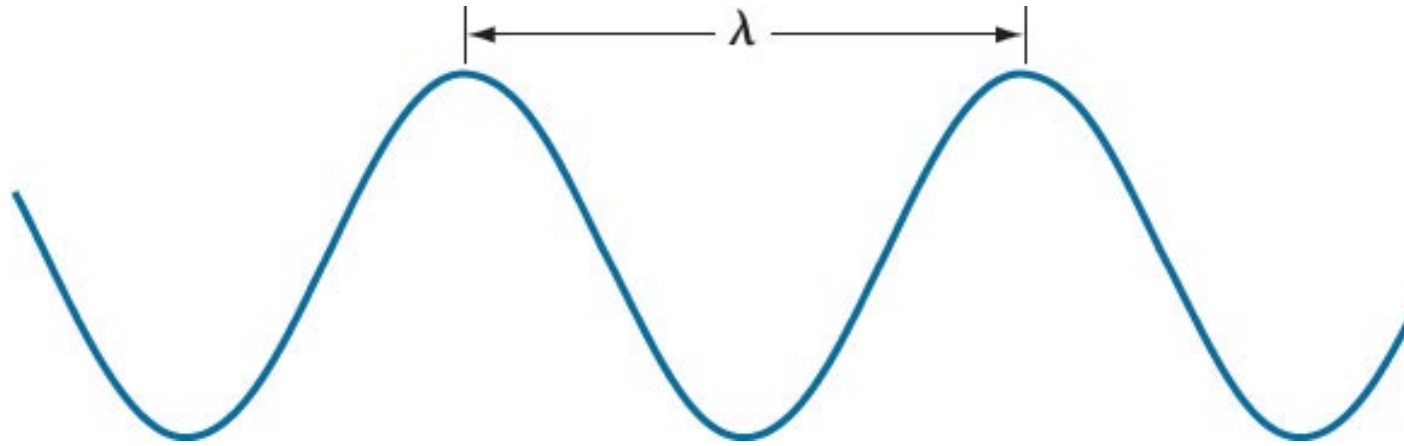


Figure 2.11

Graphical representation of one wavelength.



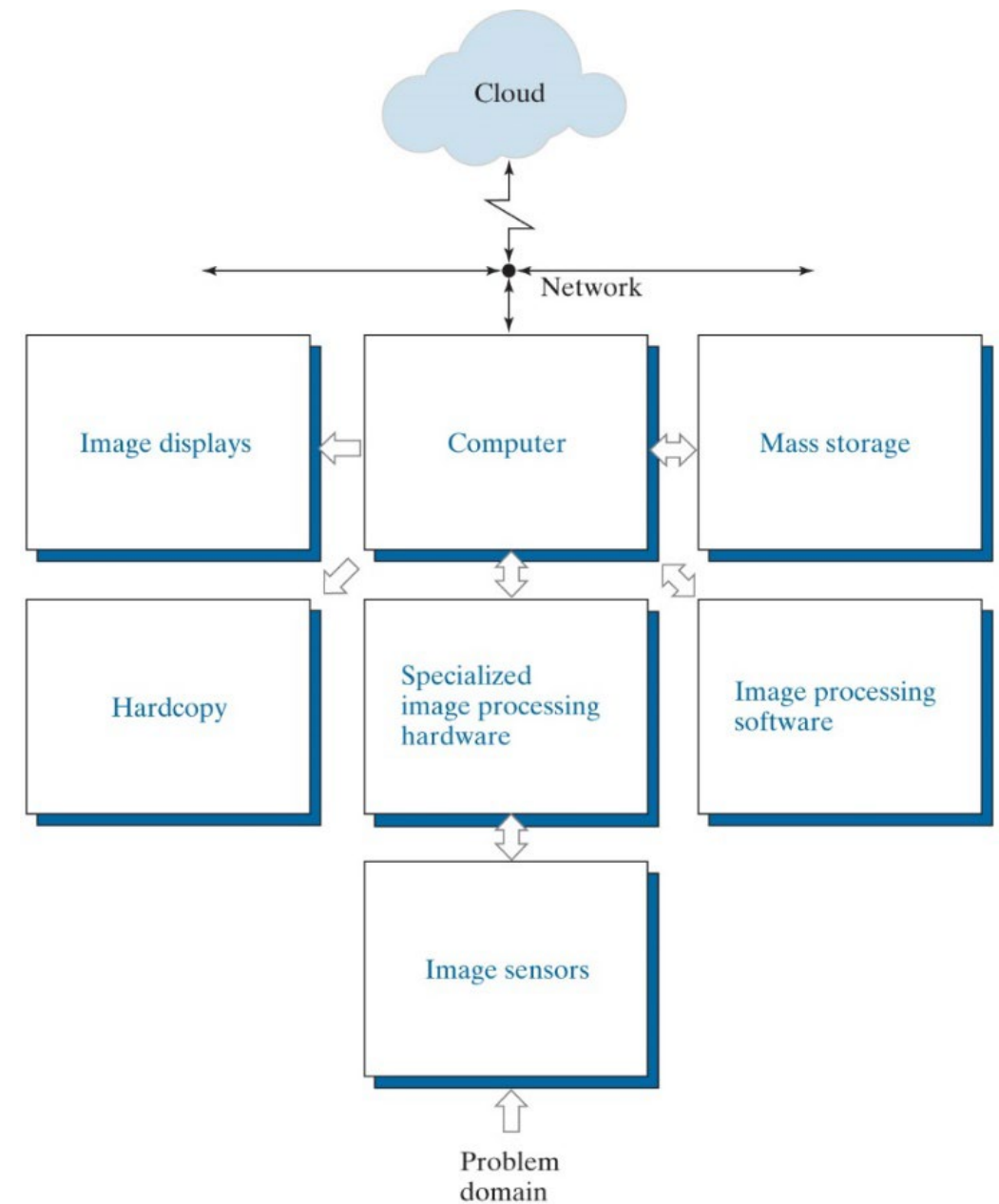
Fields using Image Processing

- Gamma-Ray Imaging
- X-Ray Imaging
- Ultraviolet band and Microscopy
- Space explorations
- Satellite images and Meteorology
- Imaging in microwave, radio bands
- MRI, ultrasound, etc.

(see section 1.3 of the book)

Figure 1.24

Components of a general-purpose image processing system.



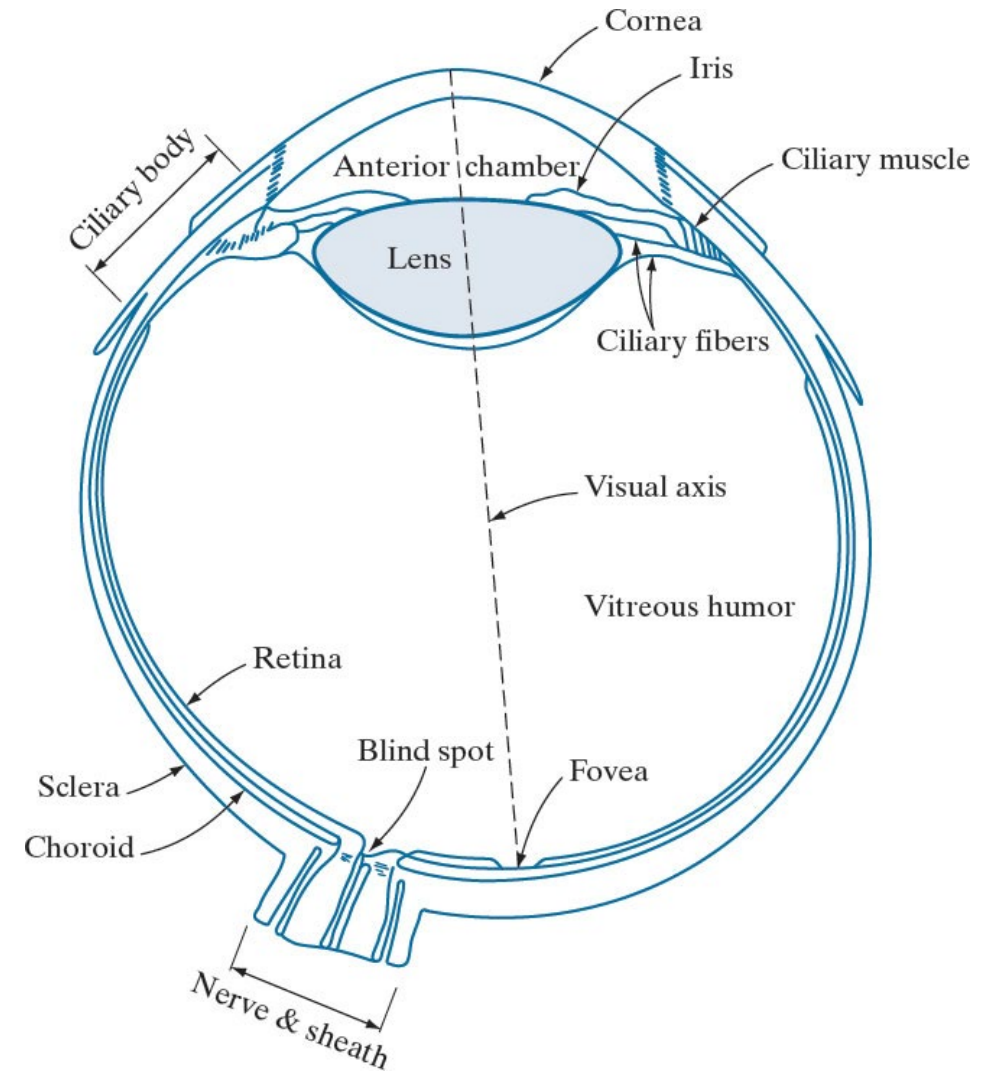
Definitions

- Image acquisition
 - Obtain an image through sensors, or read an image file; possibly some preprocessing such as scaling
- Image enhancement
 - Process of manipulating an image so the result is more suitable than the original for a *specific* application; based on human *subjective* preferences
- Image restoration
 - Improving the appearance of an image; based on *objective* mathematical or probabilistic models of image degradation.
- Color image processing
 - Color models and basic color processing in a digital domain. Color is used also as the basis for extracting features of interest in an image.
- Compression
 - Techniques for reducing the storage required to save an image, or the bandwidth required to transmit it.
- Morphological Processing
 - Tools for extracting image components that are useful in the representation and description of shape.

Human Vision System

Figure 2.1

Simplified diagram of a cross section of the human eye.



Human Eye

- Lens
 - Infrared and ultraviolet, as well as some of visible light spectrum is absorbed by the proteins within the lens.
- Retina
 - **Cones** in the fovea; focused image; sensitive to color; photopic or bright-light vision
 - **Rods** larger area of retina; overall image; sensitive to low illumination; scotopic or dim-light vision
- Blind spot
 - Optic nerve; no receptors

Figure 2.2

Distribution of rods and cones in the retina.

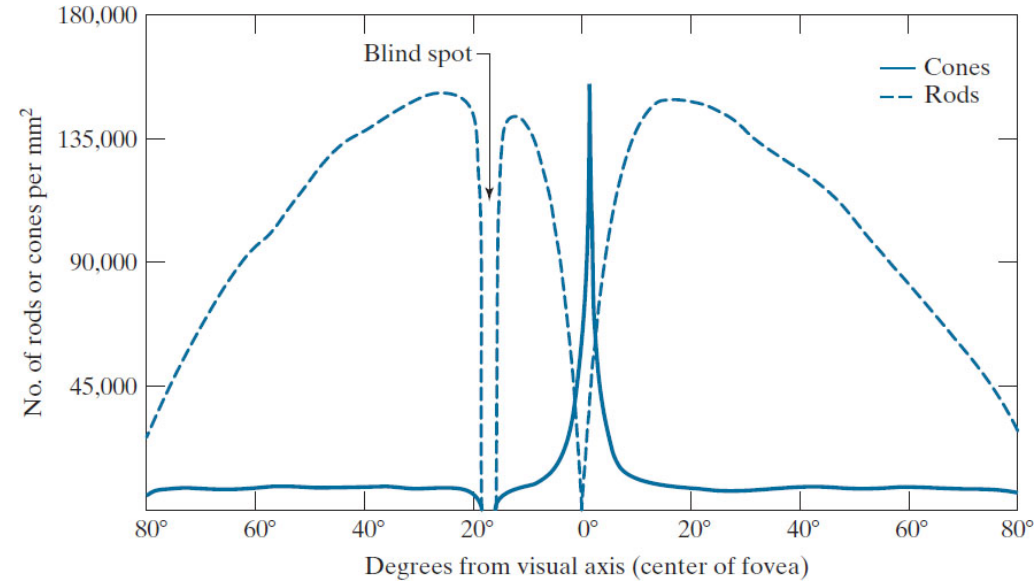
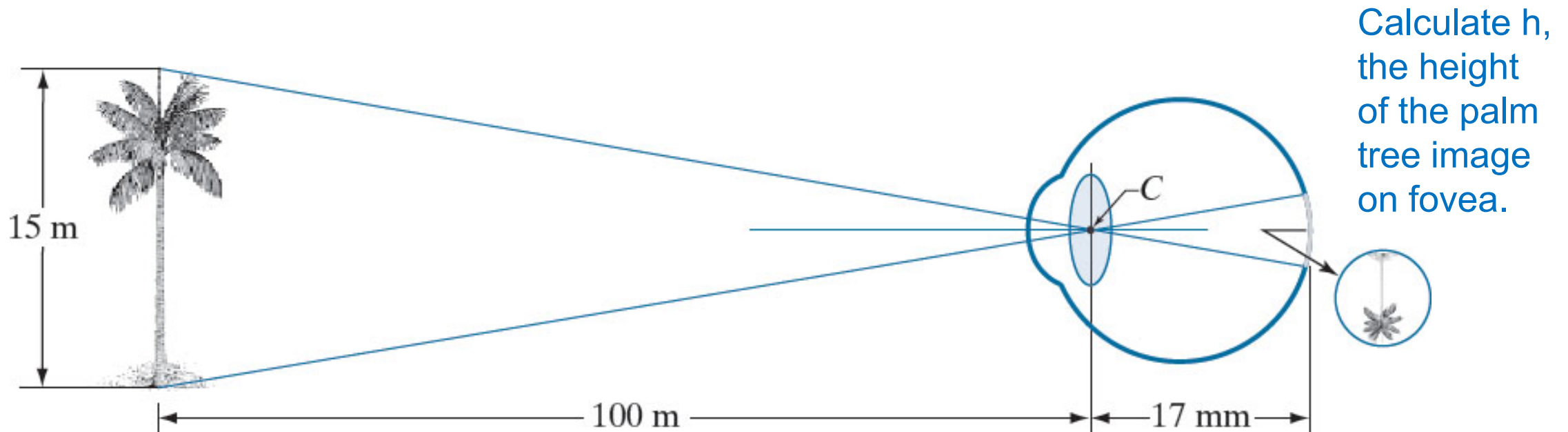


Figure 2.3

Graphical representation of the eye looking at a palm tree.
Point **C** is the focal center of the lens.



Brightness adaptation

- Human visual system can adapt to an enormous range of light intensity: from the scotopic threshold to the glare limit.
- However, NOT simultaneously! The overall sensitivity changes, a phenomenon known as brightness adaptation.
- **Brightness adaptation level:** The current sensitivity level (B_a)
- ***Subjective* brightness Range:** If the current sensitivity level is at B_a , the eye can perceive only a small range around this level.

Figure 2.4

Range of subjective brightness sensations showing a particular adaptation level, B_a .

- **Brightness adaptation level:** The current sensitivity level (B_a)
- **Subjective brightness Range:** The short intersecting curve representing the range that the eye can perceive when adapted at B_a .
 - B_b : A brightness level below which all stimuli are perceived as indistinguishable blacks.

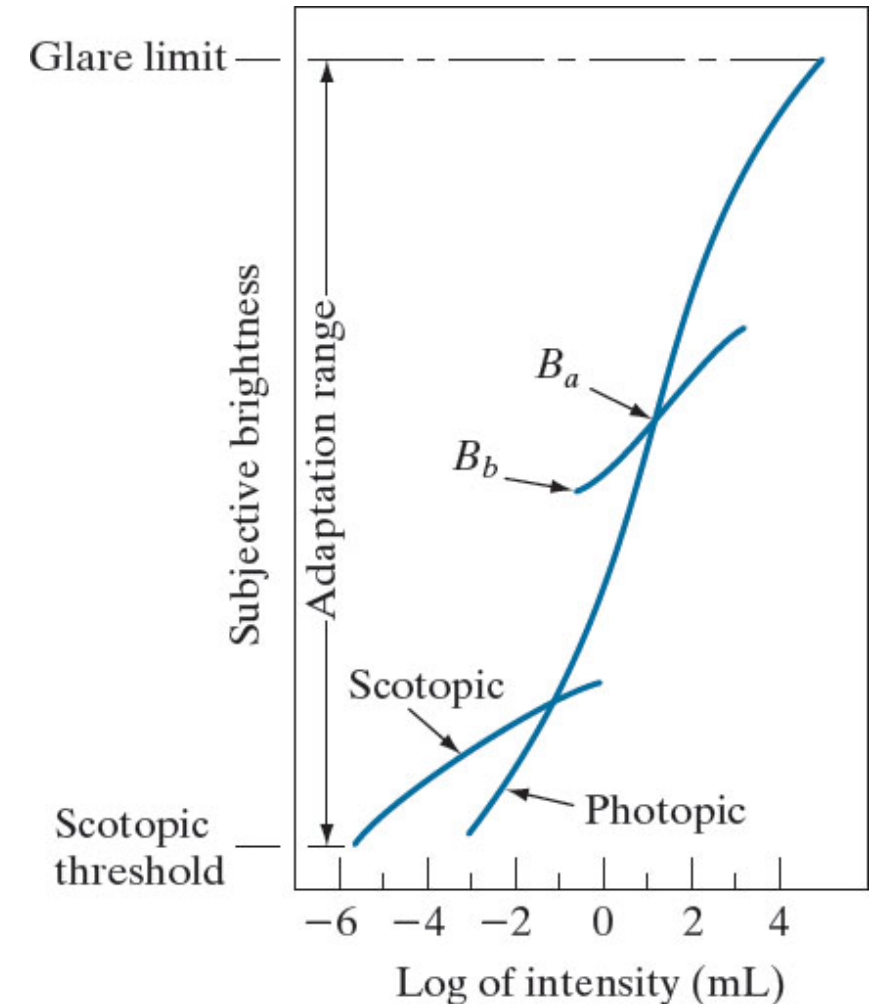


Figure 2.5

- Basic experimental setup used to characterize **brightness discrimination**.
- **Weber ratio**: The quantity $\Delta I_c / I$ where ΔI_c is the increment of illumination discriminable 50% of the time with background illumination I .

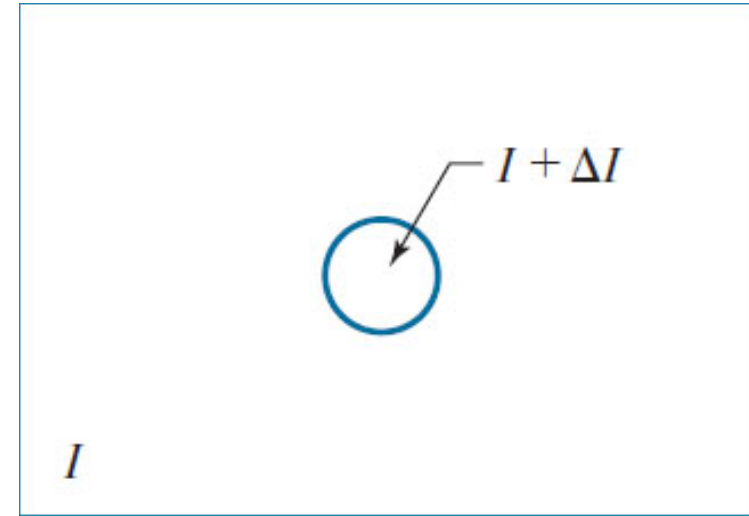


Figure 2.6

- A typical plot of the Weber ratio as a function of intensity.
- Brightness discrimination is poor (large Weber ratio) at low levels of illumination. (Rods)
- Brightness discrimination improves (low Weber ratio) at high background illumination. (Cones)

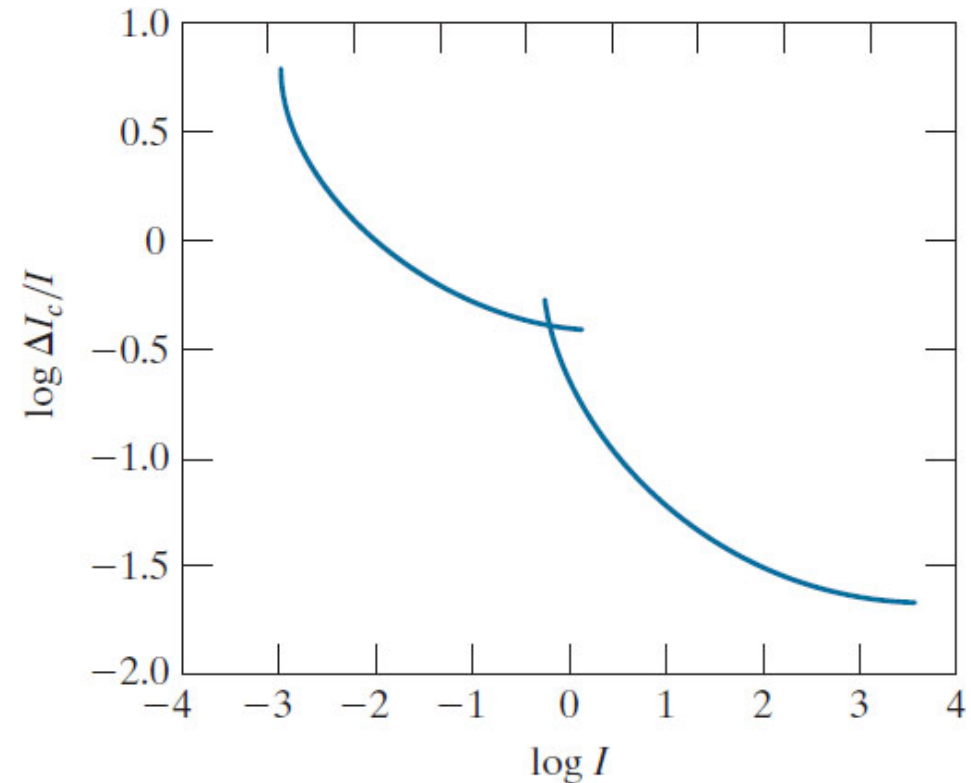


Figure 2.7

Illustration of the Mach band effect. **Perceived intensity is not a simple function of actual intensity.**

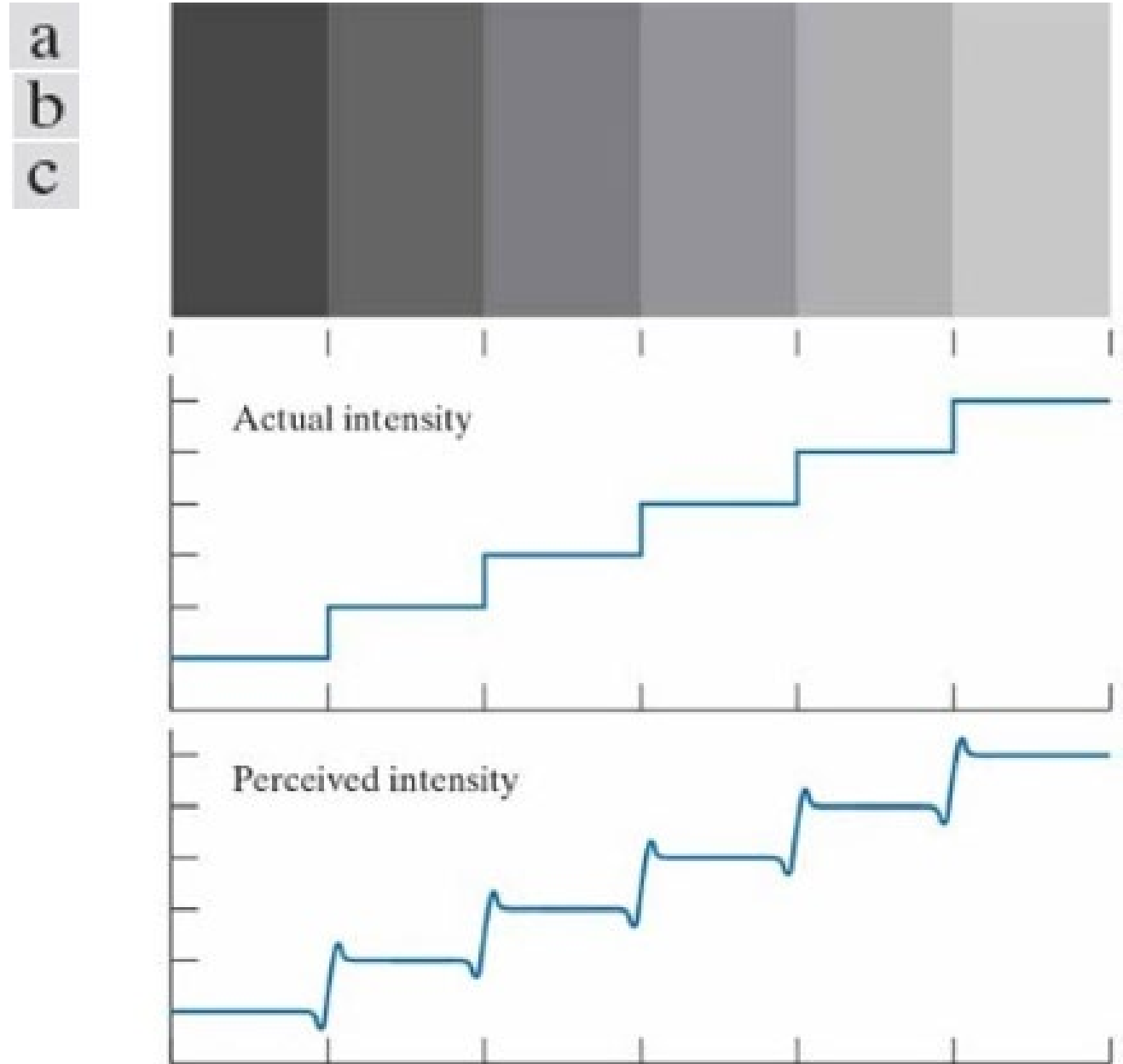


Figure 2.8

Examples of **simultaneous contrast**. All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.



Figure 2.9

Some well-known **optical illusions**.

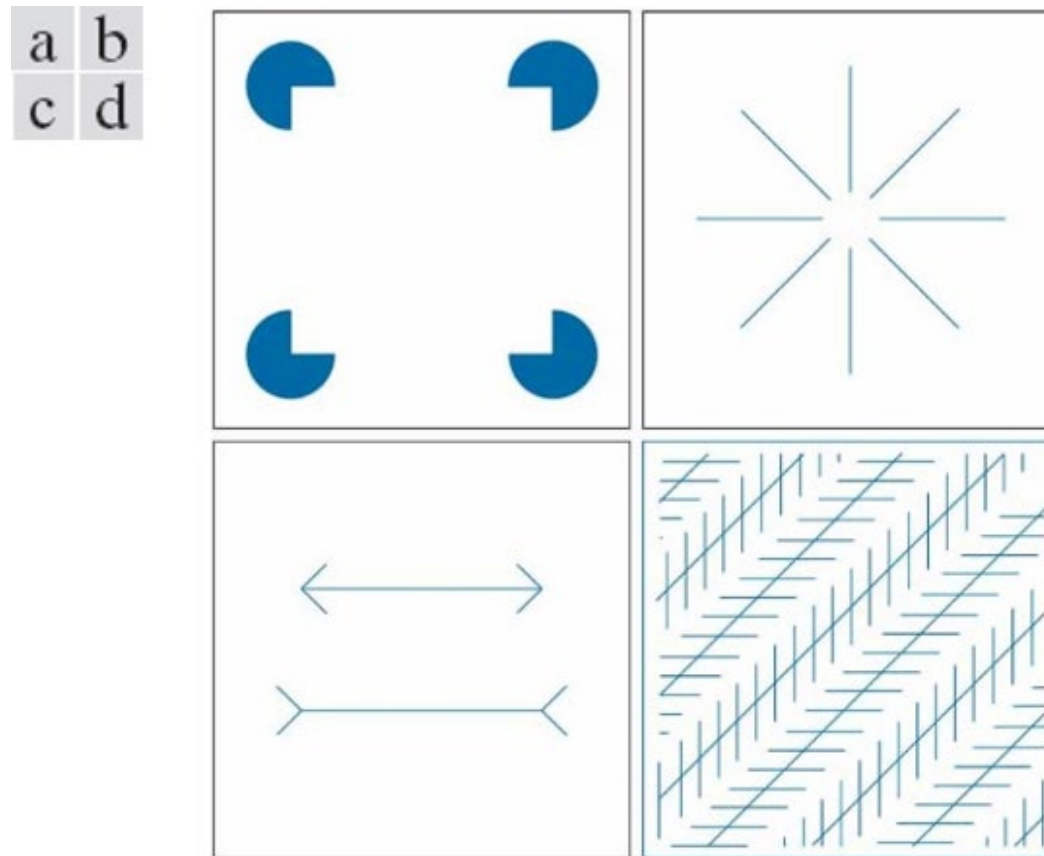


Image Acquisition and Digital Images

Figure 2.12

(a) Single sensing element. (b) Line sensor. (c) Array sensor.

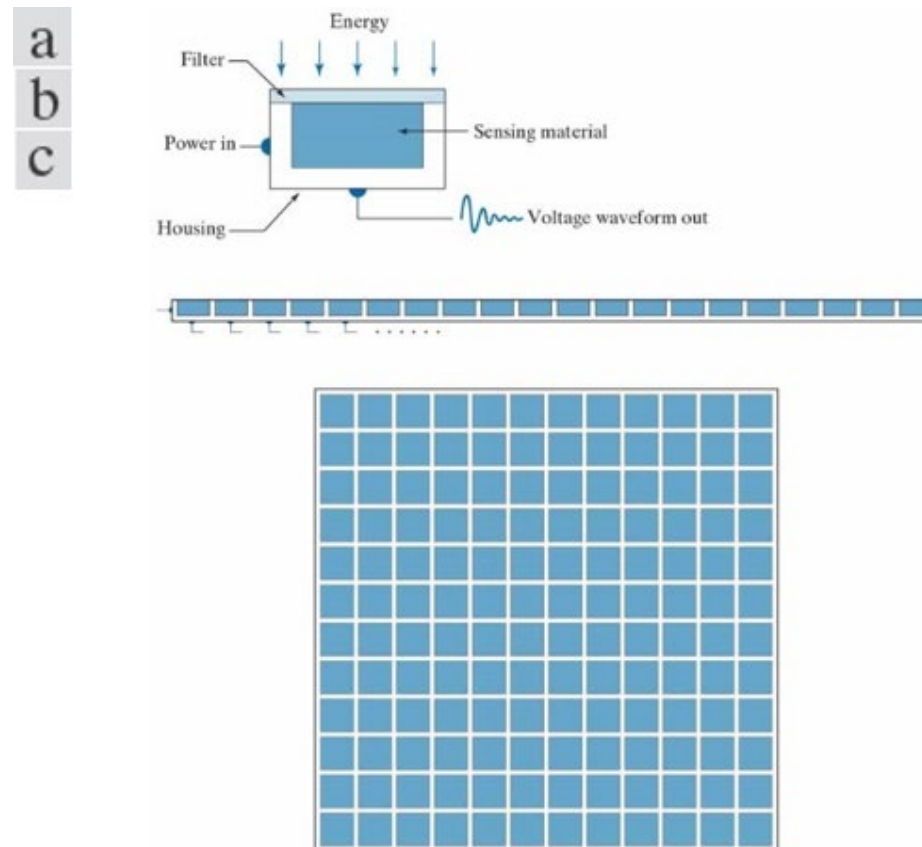


Figure 2.13

Combining a single sensing element with mechanical motion to generate a 2-D image (A light source is contained inside the drum).

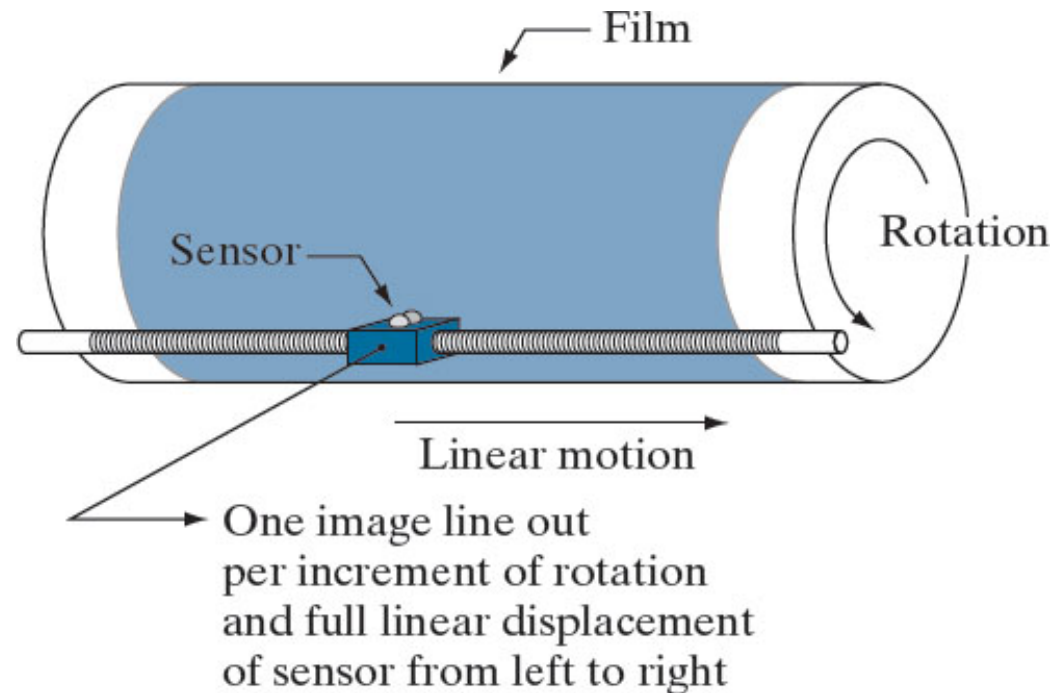


Figure 2.14

(a) Image acquisition using a linear sensor strip. (b) Image acquisition using a circular sensor strip.

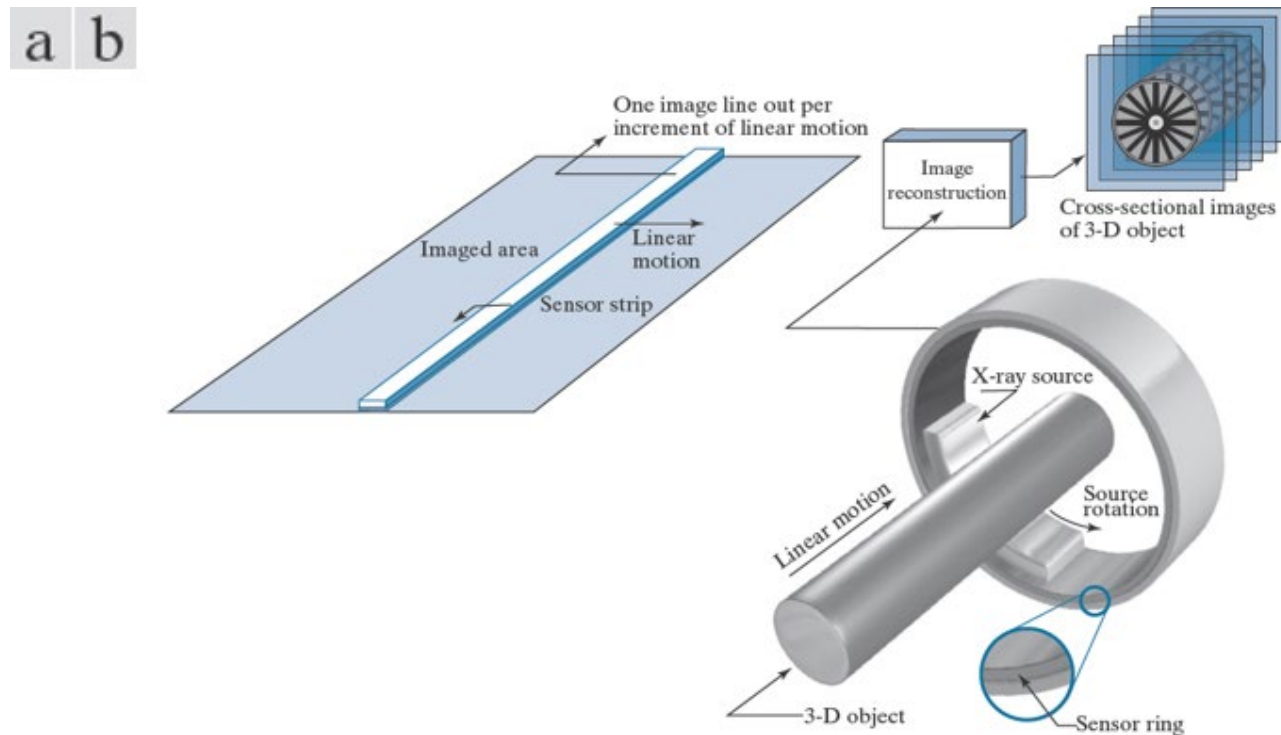
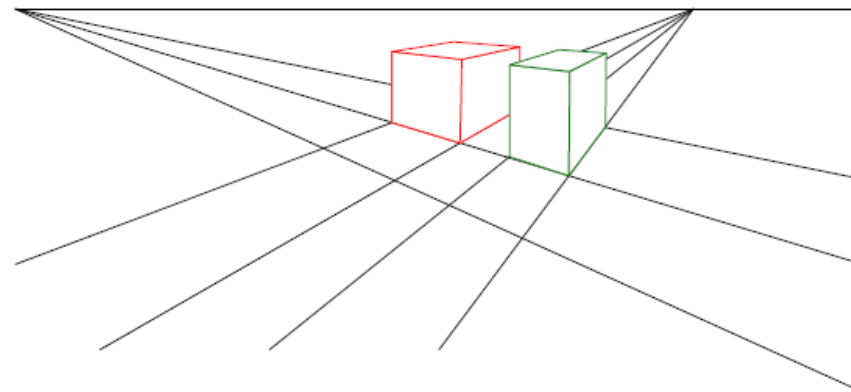
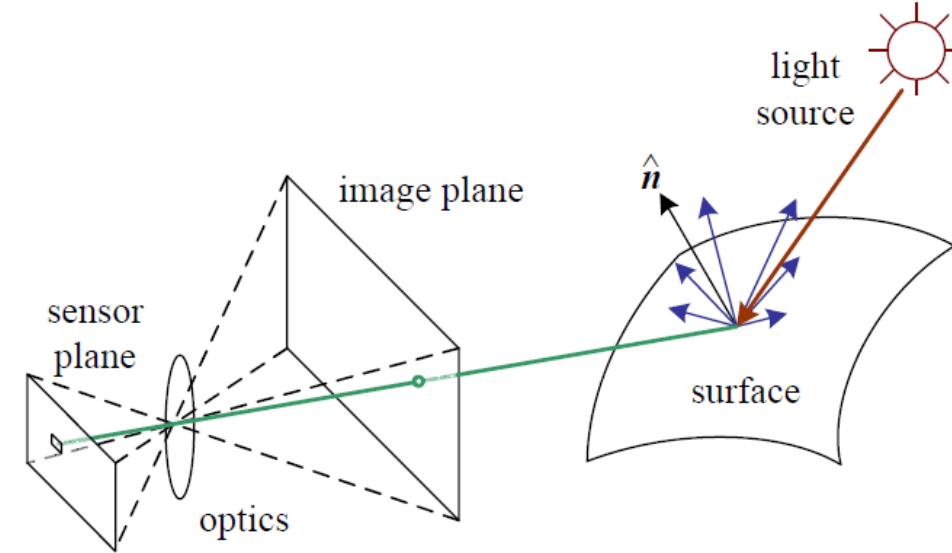


Image formation [CV]

- Simplified model of photometric image formation
 - One (or more) light sources emit light
 - Light is reflected from an object's surface in different directions
 - A small portion of reflected light enters the camera
 - An image is formed on the sensor
- 3D properties of the object are transformed into 2D image features by *Perspective Projection*.



Optics

- Assuming the lens is an ideal pinhole
 - Focal length: f
 - Aperture diameter: d
- The light from a plane at distance z_o in front of the lens, is focused onto a plane at distance z_i behind the lens

$$\frac{1}{z_o} + \frac{1}{z_i} = \frac{1}{f}$$

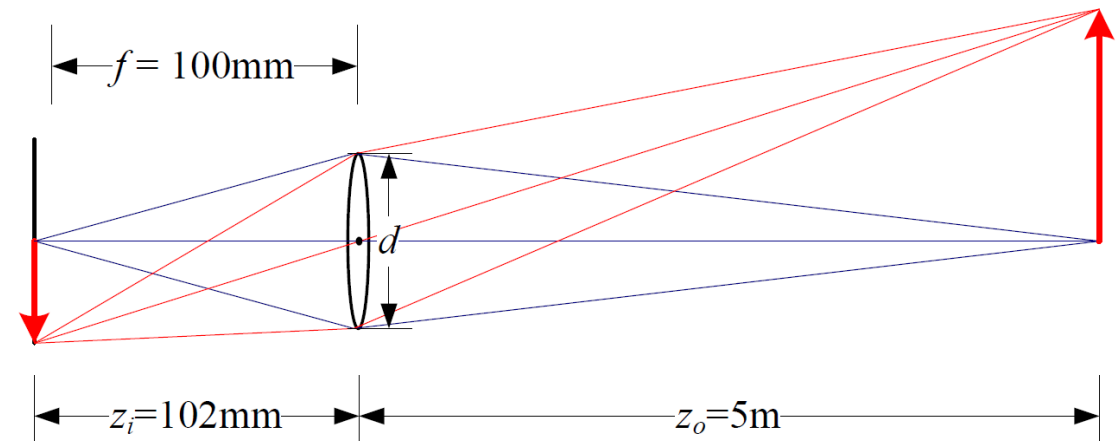
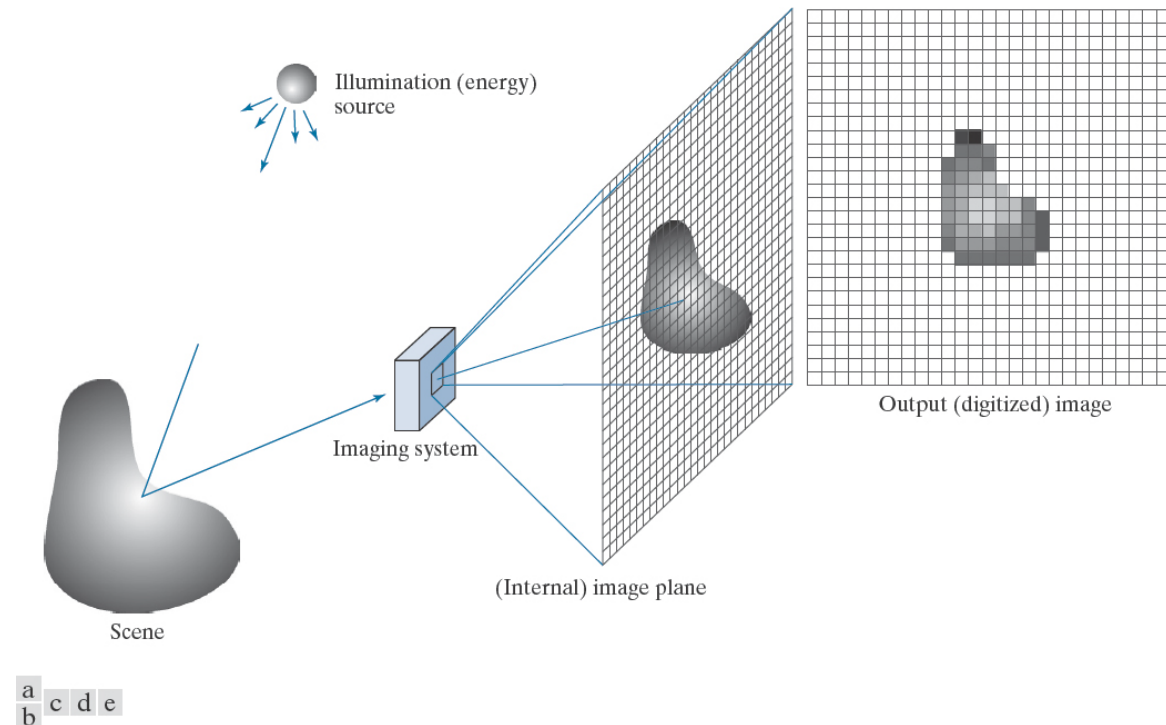


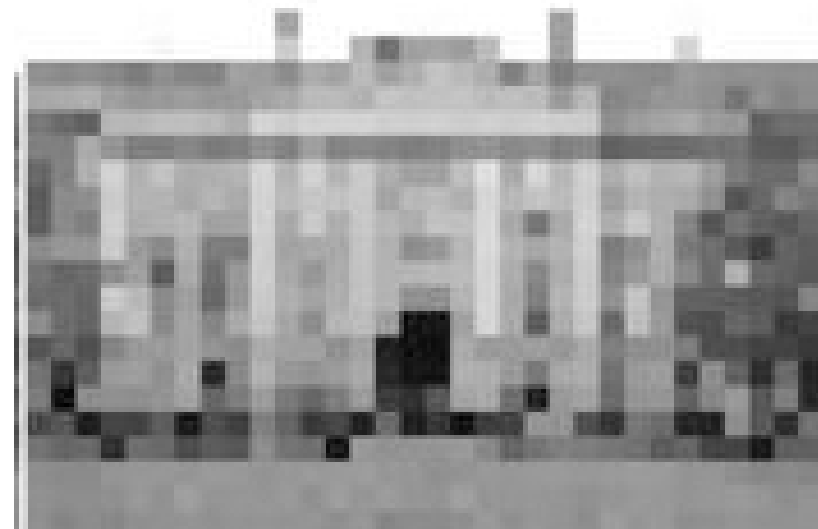
Figure 2.15

Figure 2.15 An example of digital image acquisition. (a) Illumination (energy) source. (b) A scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.



Sampling [PICV]

- Sampling
 - Samples of a continuous image into discrete elements
 - **Spatial Resolution** (number of elements)



Quantization [PICV]

- Quantization
 - Samples of continuous brightness values into discrete digital values
 - If b is the number of bits (often 8), the number of possible brightness levels is 2^b
 - **Intensity Resolution**



Figure 2.16

(a) Continuous image. (b) A scan line showing intensity variations along line **AB** in the continuous image. (c) Sampling and quantization. (d) Digital scan line. (The black border in (a) is included for clarity. It is not part of the image).

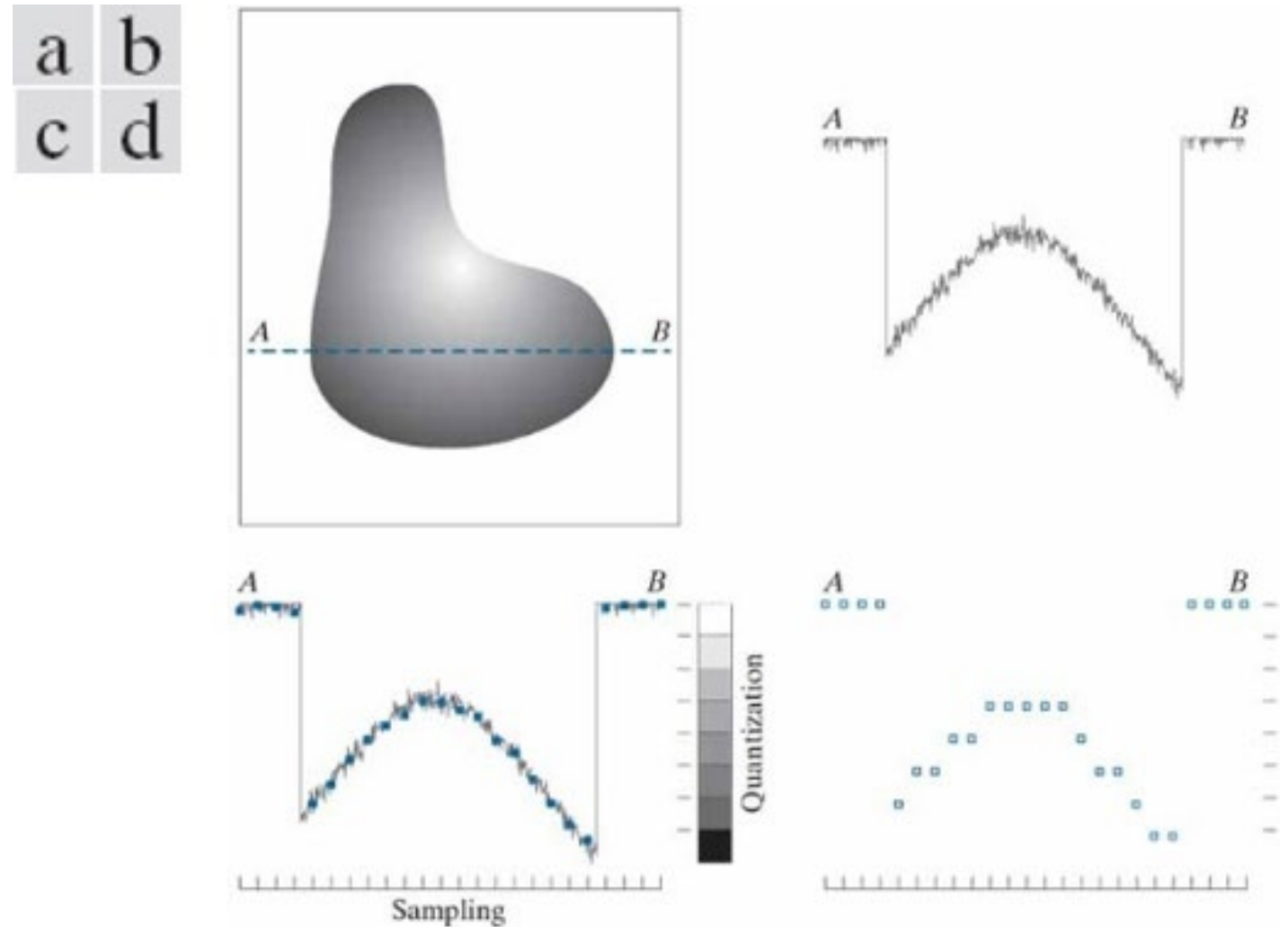


Figure 2.17

(a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

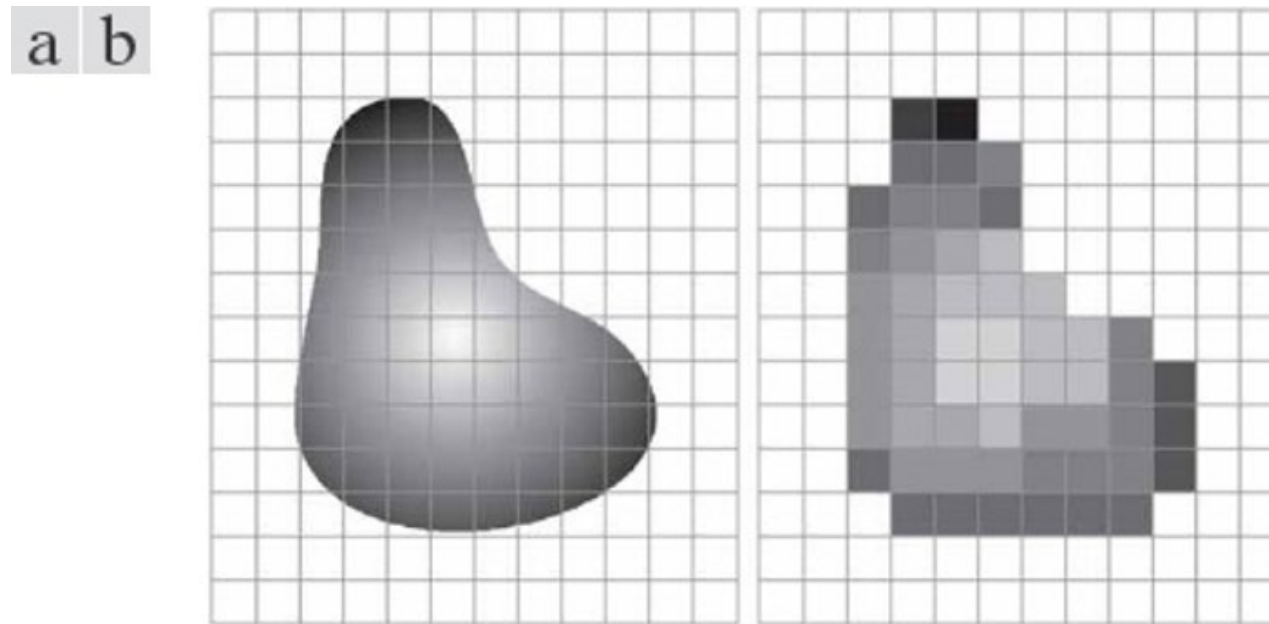


Figure 2.23

Effects of reducing spatial resolution. The images shown are at: (a) 930 dpi, (b) 300 dpi, (c) 150 dpi, and (d) 72 dpi.

a	b
c	d



Figure 2.24 (1 of 2)

(a) 2022 × 1800, 256-level image.

(b)-(d) Image displayed in 128, 64, and 32 intensity levels, while keeping the image size constant. (Original image courtesy of the National Cancer Institute.)

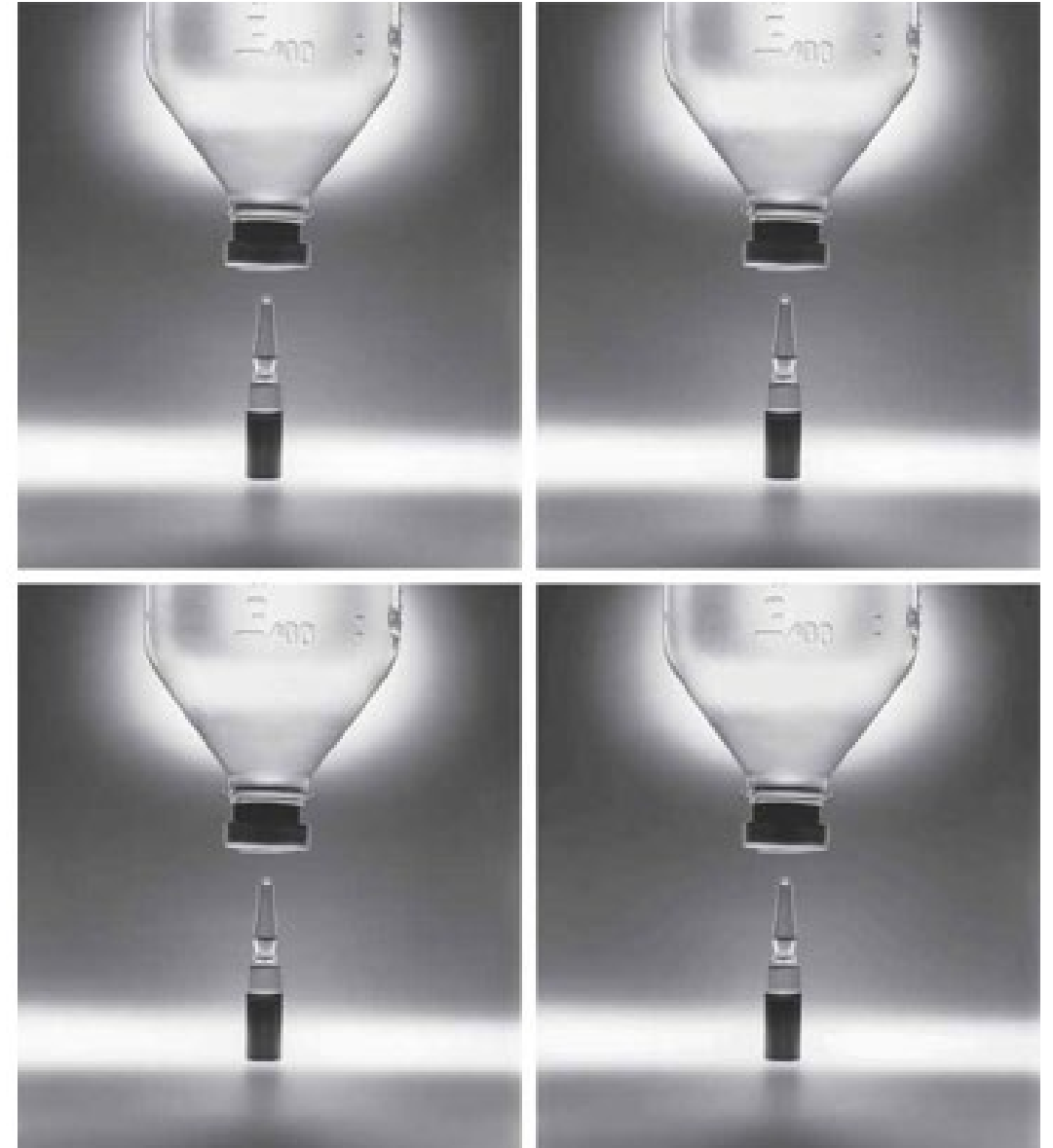
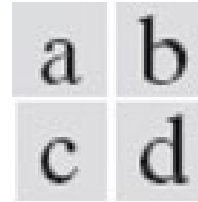


Figure 2.24 (2 of 2)

(Continued) (e)-(h) Image displayed in 16, 8, 4, and 2 intensity levels. (Original image courtesy of the National Cancer Institute.)

e	f
g	h

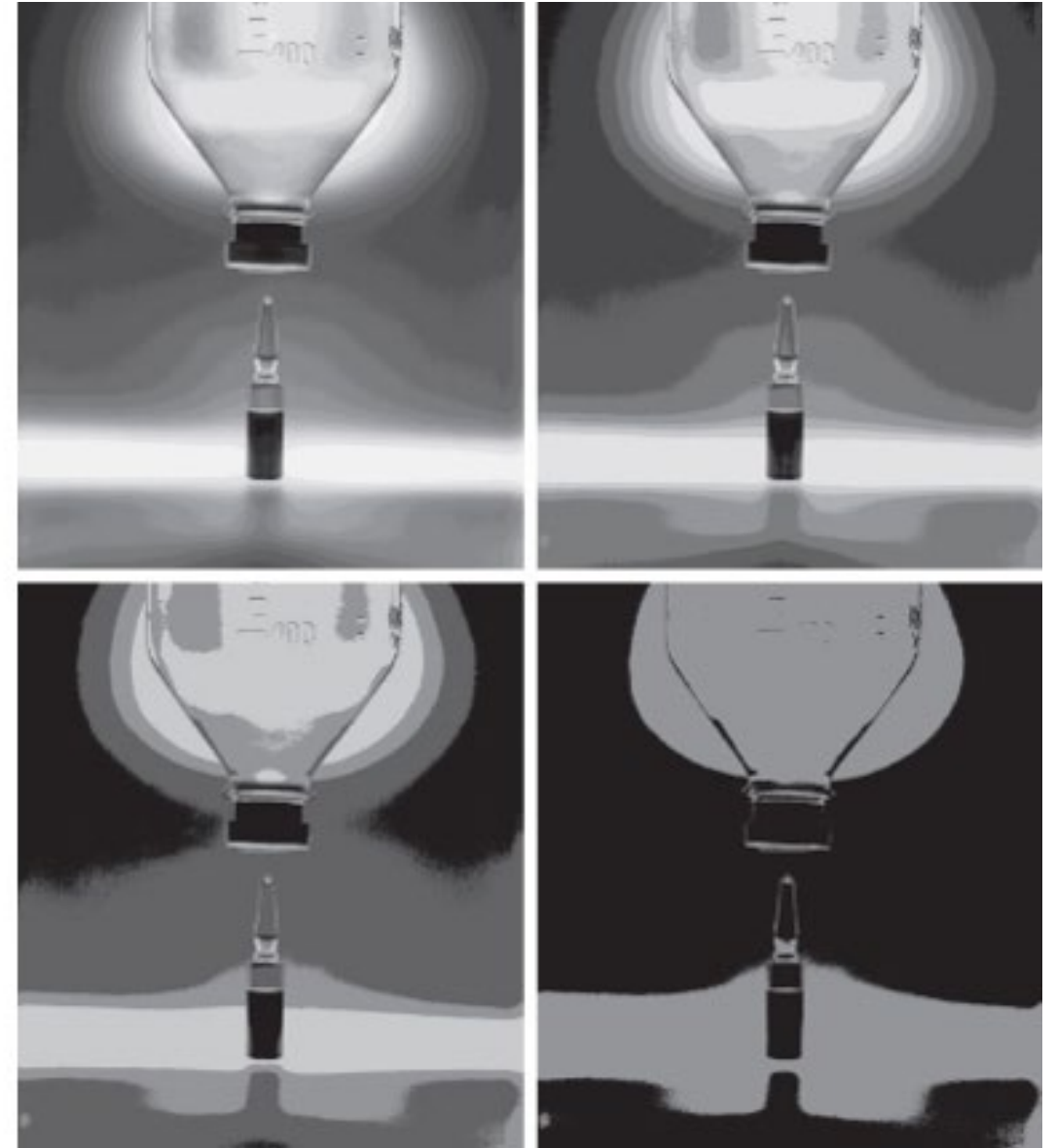


Figure 2.18

(a) Image plotted as a surface. (b) Image displayed as a visual intensity array. (c) Image shown as a 2-D numerical array. (The numbers 0, .5, and 1 represent black, gray, and white, respectively.)

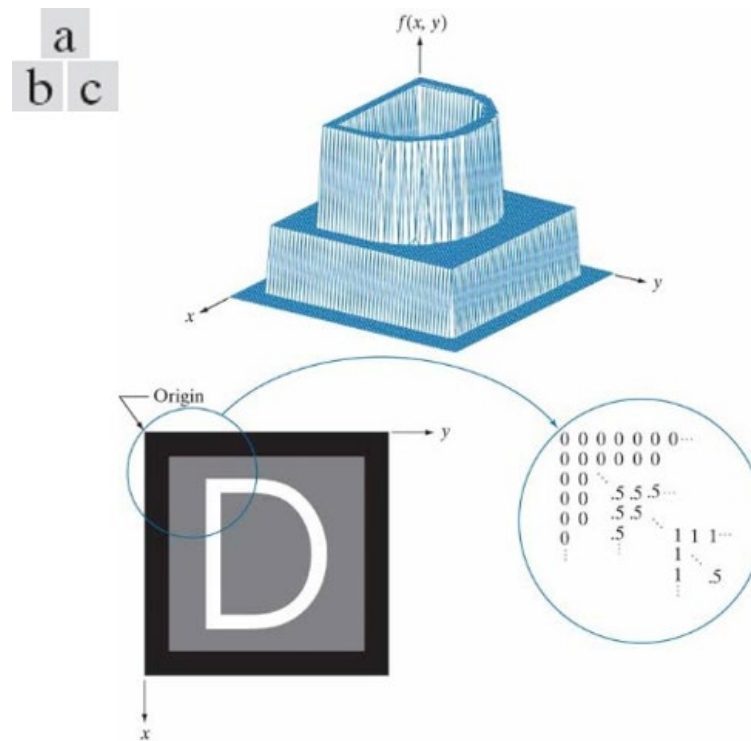


Figure 2.19

Coordinate convention used to represent digital images. Because coordinate values are integers, there is a one-to-one correspondence between $\mathbf{x}$ and $\mathbf{y}$ and the rows ($\mathbf{r}$) and columns ($\mathbf{c}$) of a matrix.

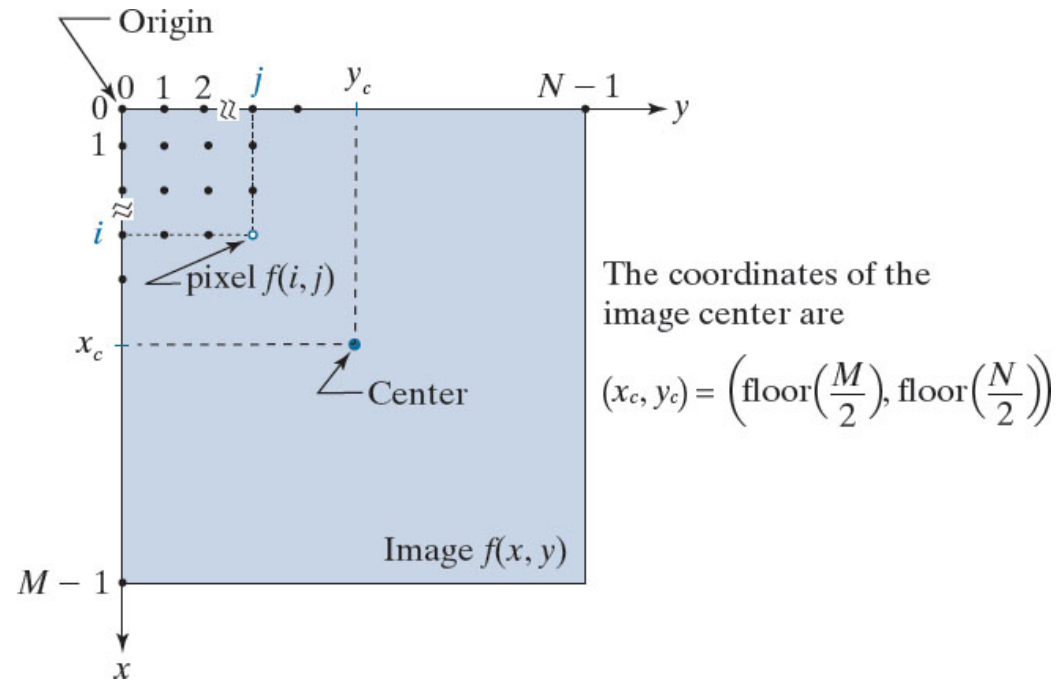
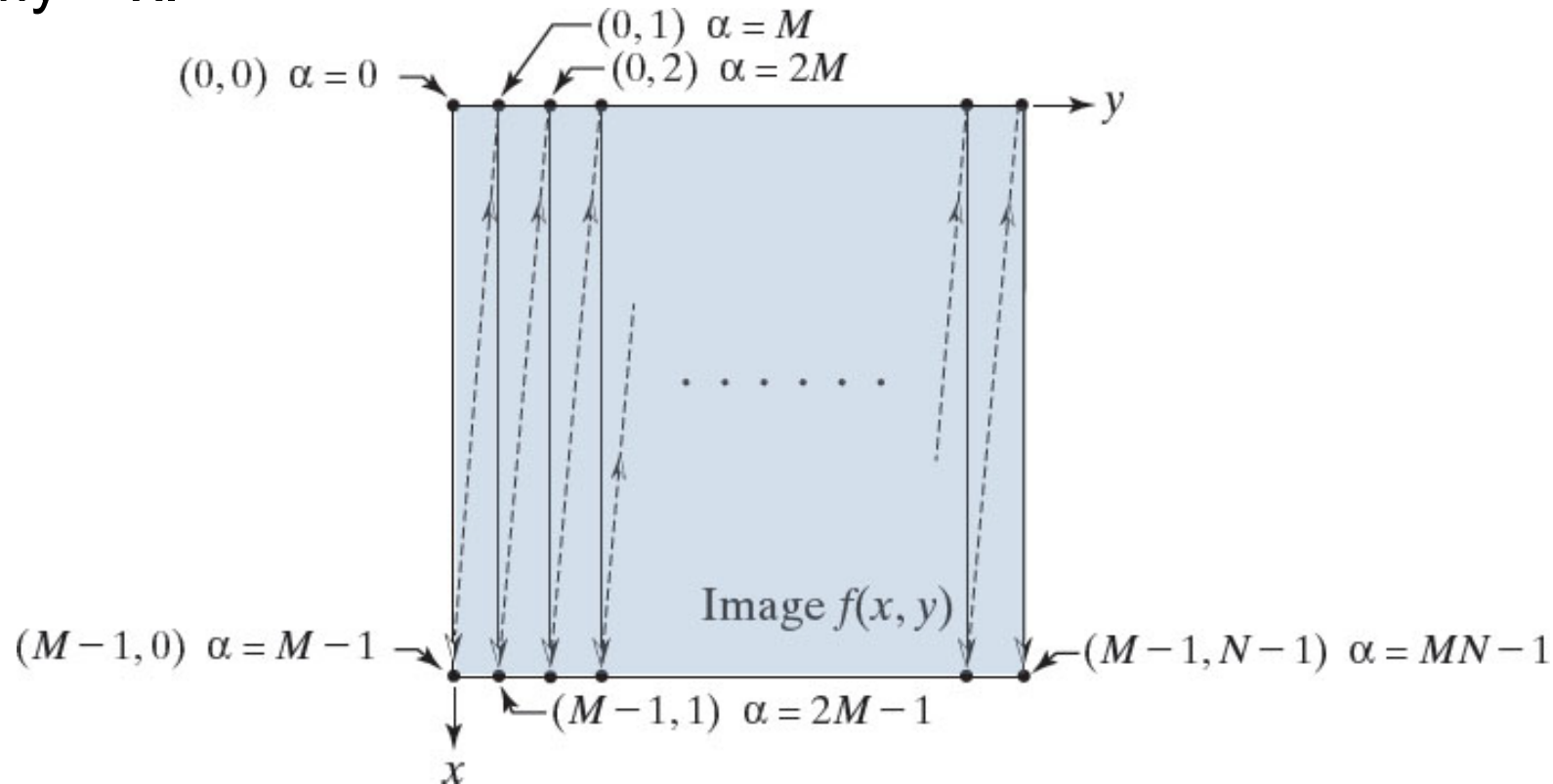


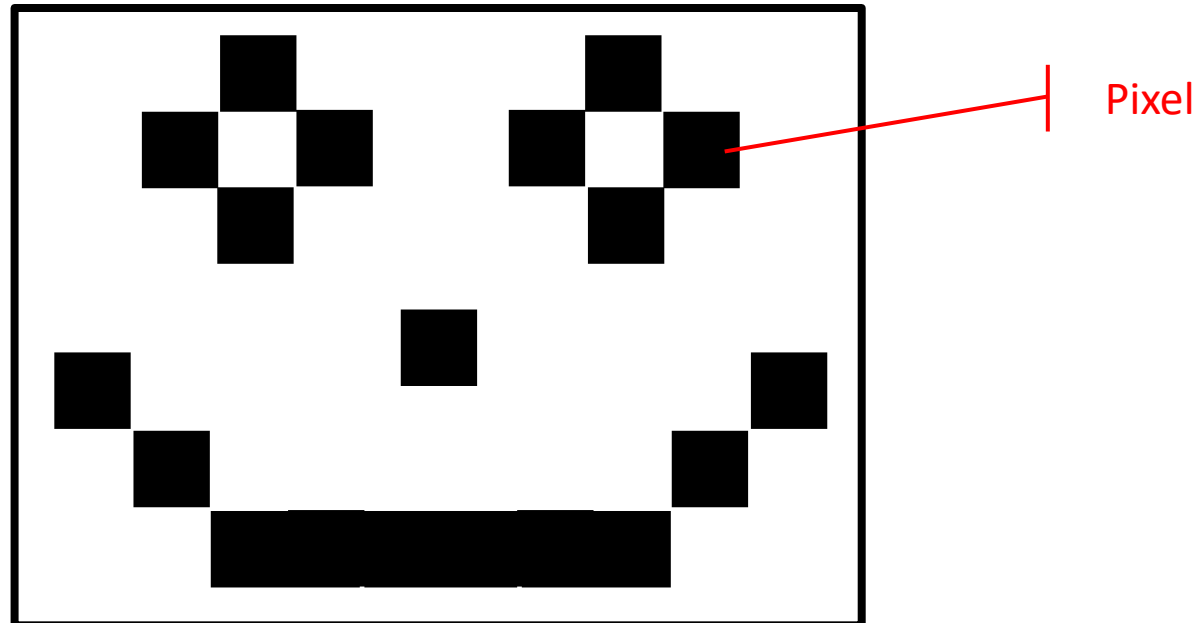
Figure 2.22

Illustration of column scanning for generating linear indices. Shown are several 2-D coordinates (in parentheses) and their corresponding linear indices $\alpha = My + x$.



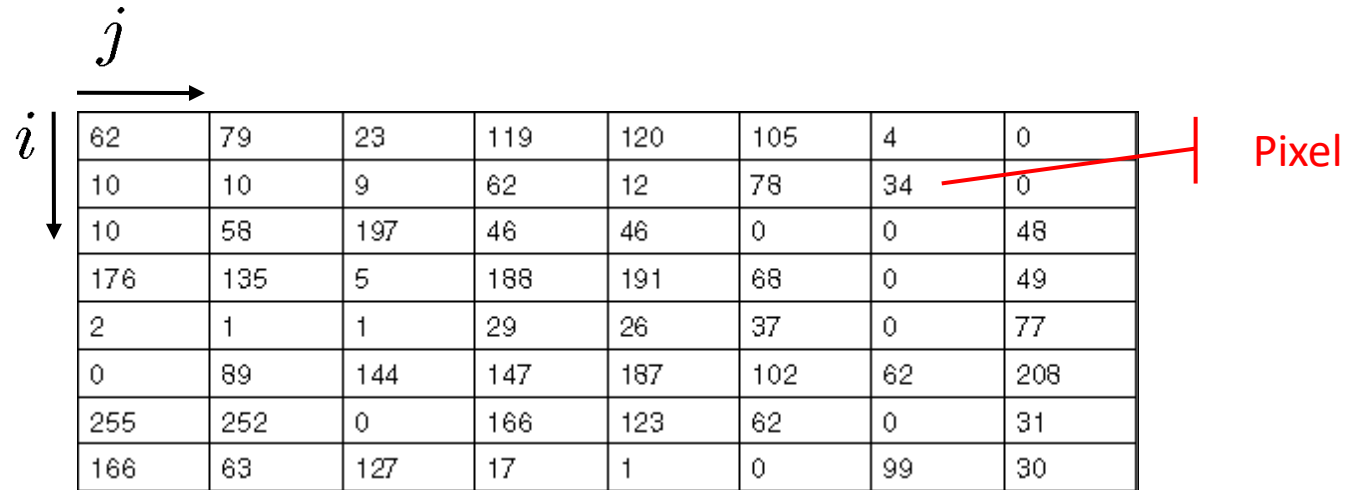
Digital images: Bitmaps and pixels

- A **bitmap** is a 2-dimensional array of bits (0's and 1's)
- Also called a **binary** image
- Each element (of this array) is called a **pixel** (picture element)



Digital images: Grayscale image

- A **grayscale** image is an array of brightness intensity values
- Often expressed in 8 bits (0 to 255)
 - Or normalized: [0, 1]



62	79	23	119	120	105	4	0
10	10	9	62	12	78	34	0
10	58	197	46	46	0	0	48
176	135	5	188	191	68	0	49
2	1	1	29	26	37	0	77
0	89	144	147	187	102	62	208
255	252	0	166	123	62	0	31
166	63	127	17	1	0	99	30

Digital images: Color and Beyond

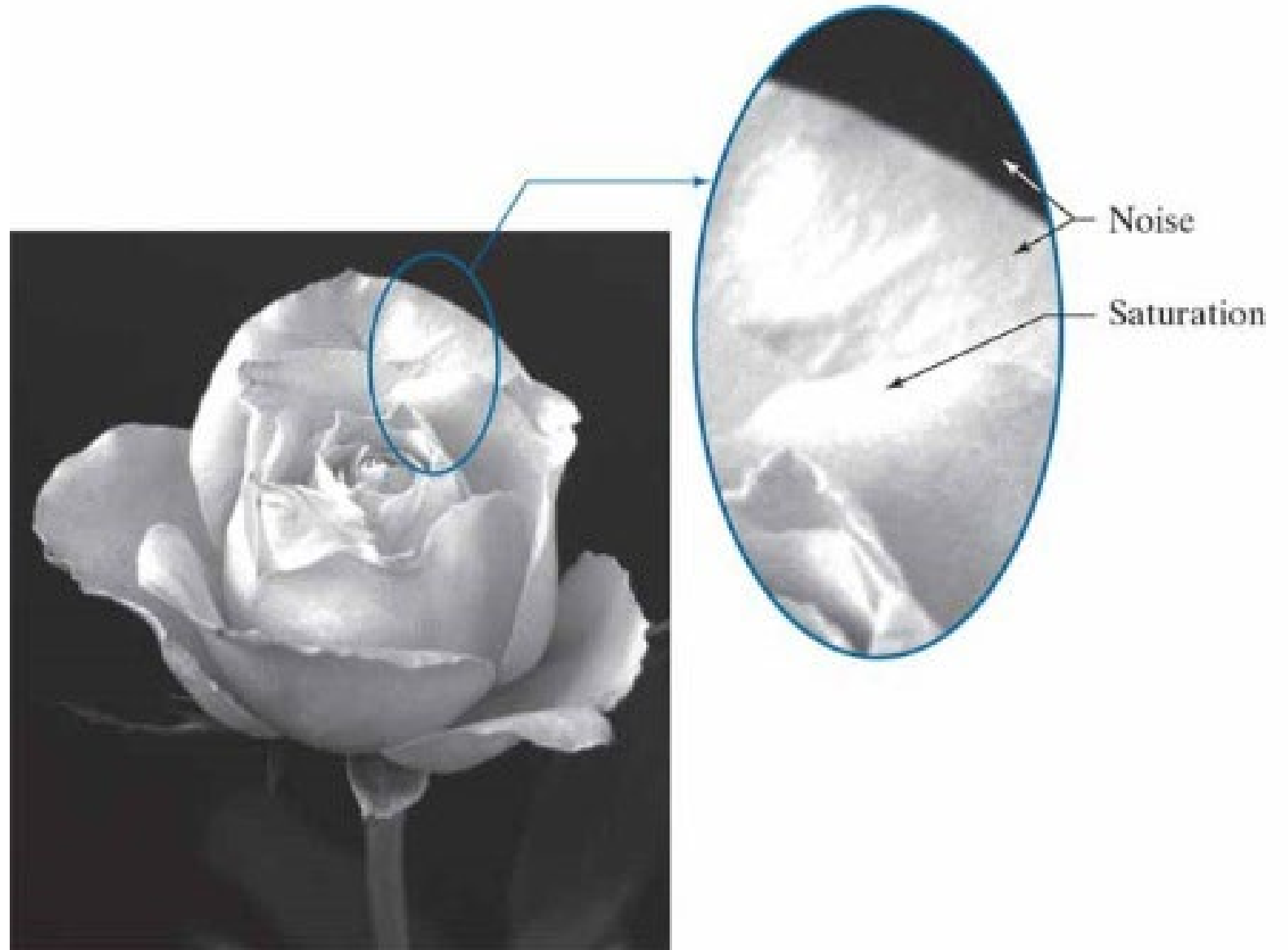
- A **color** image consists of three color maps
- Each color map is a 2D array, therefore a color image is a 3D array
- Multispectral images: may include frequencies beyond visible light spectrum
- Depth maps
 - Depth sensors, Kinect
 - Lidar (light and radar?)
 - Calculated using stereo, motion, shadows, structured light, etc.

62	79	23	119	120	105	4	0		
10	62	79	23	119	120	105	4	0	
10	10	62	79	23	119	120	105	4	0
176	10	10	10	9	62	12	78	34	0
2	176	10	58	197	46	46	0	0	48
0	2	176	135	5	188	191	68	0	49
255	0	2	1	1	29	26	37	0	77
166	255	0	89	144	147	187	102	62	208
	166	255	252	0	166	123	62	0	31
		166	63	127	17	1	0	99	30

Figure 2.20

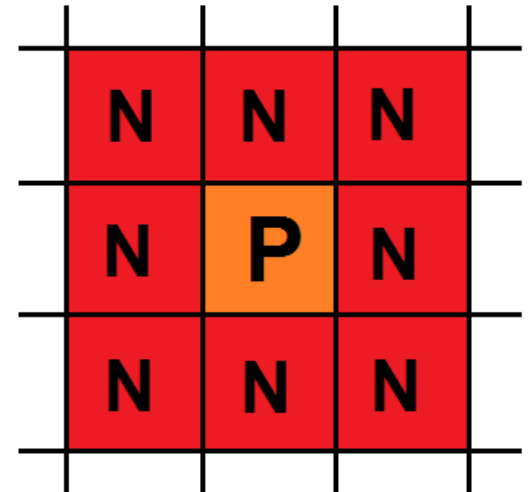
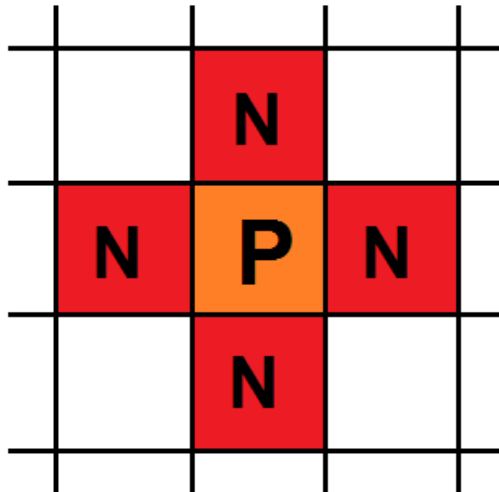
An image exhibiting saturation and noise.

Saturation is the highest value beyond which all intensity values are clipped (note how the entire saturated area has a high, constant intensity level). Visible **noise** in this case appears as a grainy texture pattern. The dark background is noisier, but the noise is difficult to see.



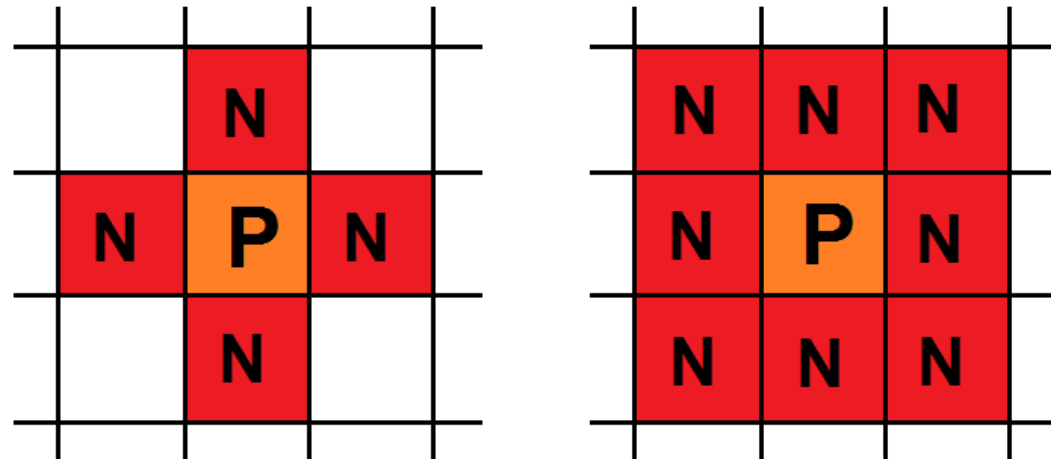
Neighbors of a pixel

- V : set of intensity values used to define adjacency.
- 4-neighbors of p , $N_4(p)$
 - $(x+1, y), (x-1, y), (x, y+1), (x, y-1)$
- Diagonal or D-neighbors of p , $N_D(p)$
 - $(x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1)$
- 8-neighbors of p , $N_8(p)$
 - $N_4(p) \cup N_D(p)$



Connectivity (or Adjacency)

- V : set of intensity values used to define adjacency.
- 4- connectivity: A pixel p is considered connected to 4 neighbors
 - Two pixels p and q with values from V are 4-adjacent if q is in the set $N_4(p)$.
- 8- connectivity: A pixel p is considered connected to 8 neighbors
 - Two pixels p and q with values from V are 8-adjacent if q is in the set $N_8(p)$.

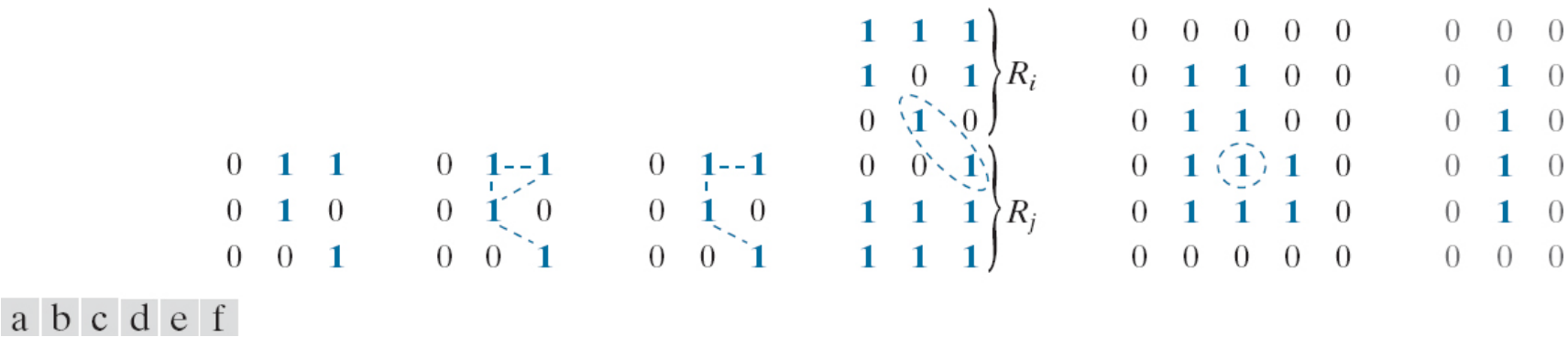


Mixed-Adjacency (m-Adjacency)

- Two pixels p and q with values from V are m-adjacent if
 - q is in the set $N_4(p)$, or
 - q is in the set $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels whose values are from V .

Figure 2.28

(a) An arrangement of pixels. (b) Pixels that are 8-adjacent (adjacency is shown by dashed lines). (c) **m**-adjacency. (d) Two regions (of 1's) that are 8-adjacent. (e) The circled point is on the boundary of the 1-valued pixels only if 8-adjacency between the region and background is used. (f) The inner boundary of the 1-valued region does not form a closed path, but its outer boundary does.



Connectivity to Regions

- A **path** (or curve) is a sequence of distinct pixels $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ where (x_i, y_i) and (x_{i-1}, y_{i-1}) are connected (4-, 8-, or m-paths).
- **Connected set or Region:** A subset S of pixels where for any two pixels p and q in S there exists a path between them consisting entirely of pixels in S .
- The **boundary** (also called the border or contour) of a region R is the set of pixels in R that are adjacent to pixels in the complement of R .
 - The above definition is also referred to as the *inner border* (vs. *outer border*).

Table 2.1

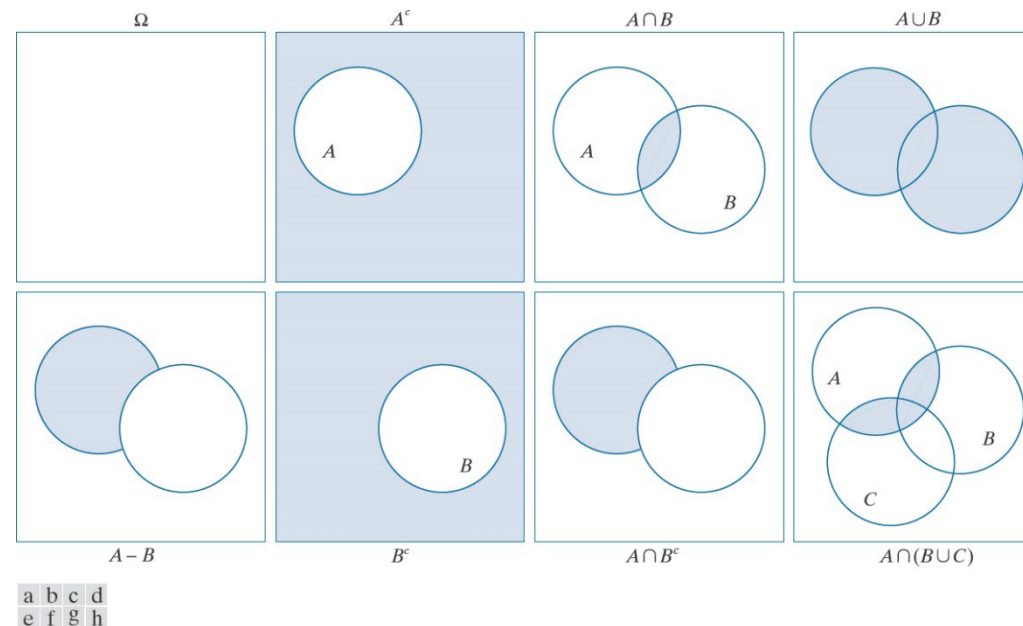
Some important set operations and relationships.

Description	Expressions
Operations between the sample space and null sets	$\Omega^c = \emptyset; \emptyset^c = \Omega; \Omega \cup \emptyset = \Omega; \Omega \cap \emptyset = \emptyset$
Union and intersection with the null and sample space sets	$A \cup \emptyset = A; A \cap \emptyset = \emptyset; A \cup \Omega = \Omega; A \cap \Omega = A$
Union and intersection of a set with itself	$A \cup A = A; A \cap A = A$
Union and intersection of a set with its complement	$A \cup A^c = \Omega; A \cap A^c = \emptyset$
Commutative laws	$A \cup B = B \cup A$ $A \cap B = B \cap A$
Associative laws	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
Distributive laws	$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$
DeMorgan's laws	$(A \cup B)^c = A^c \cap B^c$ $(A \cap B)^c = A^c \cup B^c$

Figure 2.35

Venn diagrams corresponding to some of the set operations in Table 2.1. The results of the operations, such as A^c ,

are shown shaded. Figures (e) and (g) are the same, proving via Venn diagrams that $A - B = A \cap B^c$ [see Eq.(2 - 40)].



Distance Measures

- A distance function or metric D , if

a. $D(p, q) \geq 0$ ($D(p, q) = 0$ iff $p = q$),

b. $D(p, q) = D(q, p)$, and

c. $D(p, s) \leq D(p, q) + D(q, s)$.

- Euclidean distance
- City-block (Manhattan) distance
- Chessboard distance

$$D_e(p, q) = \left[(x - u)^2 + (y - v)^2 \right]^{\frac{1}{2}}$$

$$D_4(p, q) = |x - u| + |y - v|$$

$$D_8(p, q) = \max(|x - u|, |y - v|)$$

Summary-1

- Image are created from various energy sources. The principal energy source for images in use today is the **electromagnetic (EM) energy spectrum**.
- Image **enhancement** is for a specific application and evaluated based on human subjective preferences, while **restoration** is based on objective mathematical or probabilistic models of image degradation.
- Photopic or bright-light vision uses cones in the fovea, while scotopic or dim-light vision uses rods. Subjective brightness range is the range of light intensity the eye can adapt to around the brightness adaptation level.
- Weber ratio is a measure used to show brightness discrimination at various levels of illumination.

Summary-2

- The optics formulas are used to find the size of image or distance from lens given the focal length.
- Two types of sampling in digital images: spatial sampling, and intensity sampling (quantization).
- Three types of adjacency are defined as: 4-adjacency, 8-adjacency, and m-adjacency.
- Several popular distance measures used in image processing are: Euclidean, Manhattan, and chessboard distances.

References

- [DIP] Gonzalez, R.C. and Woods, R.E., Digital Image Processing, Pearson, 4th ed., 2018.
- [CV] Computer Vision: Algorithms and Applications, R. Szeliski (<http://szeliski.org/Book>)
- [PICV] Practical introduction to Computer Vision with OpenCV, Kenneth Dawson-Howe, WILEY, 2014. Available through Seneca libraries

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